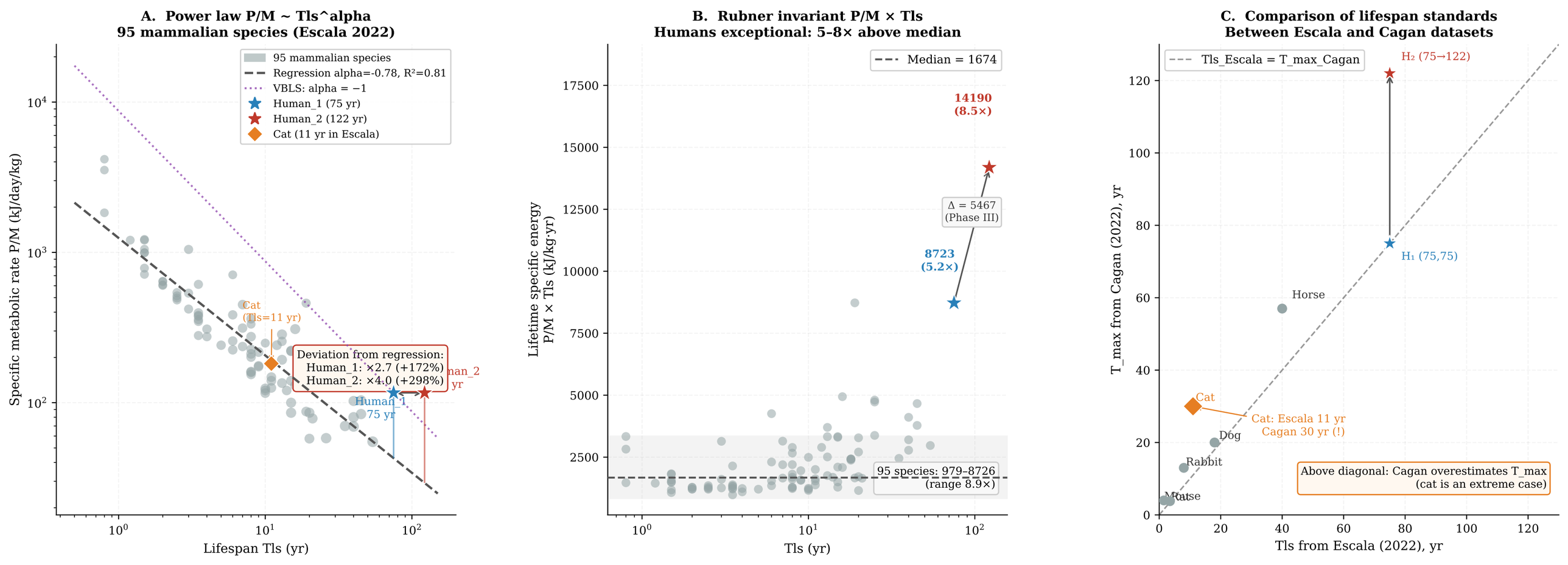

A. Power law P/M ~ Tls^alpha
95 mammalian species (Escala 2022)
95 mammalian species
Regression alpha=-0.78, R²=0.81
VBLS: alpha = −1
Human_1 (75 yr)
Human_2 (122 yr)
Cat (11 yr in Escala)
Specific metabolic rate P/M (kJ/day/kg)
Lifespan Tls (yr)
Cat
(Tls=11 yr)
Deviation from regression:
Human_1: ×2.7 (+172%)
Human_2: ×4.0 (+298%)
Human_1
75 yr
B. Rubner invariant P/M × Tls
Humans exceptional: 5–8× above median
Median = 1674
14190
(8.5×)
Δ = 5467
(Phase III)
8723
(5.2×)
95 species: 979–8726
(range 8.9×)
Lifetime specific energy
P/M × Tls (kJ/kg·yr)
Tls (yr)
C. Comparison of lifespan standards
Between Escala and Cagan datasets
Tls_Escala = T_max_Cagan
H₂ (75→122)
H₁ (75,75)
Horse
Cat
Cat: Escala 11 yr
Cagan 30 yr (!)
Dog
Rabbit
Mouse
Above diagonal: Cagan overestimates T_max
(cat is an extreme case)
T_max from Cagan (2022), yr
Tls from Escala (2022), yr